

Designing Anti-Tautological Probes on the 6N Skeleton:

A Negative Case Study with Elliptic-Curve Ranks and the CRT Information Barrier

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6N Arithmetic Geodynamics series · Methodological Note · 2026

Abstract

This note is methodological, not a claim of new arithmetic. Within the heuristic “6N” framework (primes >3 lie on the residue classes $6N \pm 1$), we ask whether the 6N wing of a prime p exerts any control over a high-dimensional arithmetic invariant attached to p : the Mordell–Weil rank of the elliptic curve $E_p : y^2 = x^3 - p$. We first articulate a discipline for distinguishing genuine probes from *tautological* ones — probes whose observable is a re-encoding of primality — and argue that the rank of E_p is the one clean, primality-independent observable with a natural 6N bridge ($p \bmod 6$). Using certified true ranks for the 1116 curves with $p < 10^4$ computable to certainty (SageMath), a three-layer analysis (raw wing effect; effect within fixed root-number classes; label-shuffle null) finds no wing control of rank: Cohen’s $d = +0.022$ raw, $d \approx +0.02$ within both root-number classes, and a shuffle test yielding $p = 0.913$. Rank is organized entirely by the root number (a function of $p \bmod 24$). We interpret this as a concrete instance of a **Chinese-Remainder information barrier**: 6N is a mod-6 datum and is, by the CRT factorisation of the relevant conductor/root-number data, absorbed by the finer mod-24 structure that governs the invariant. The contribution is the falsification discipline and a worked, honestly-negative example — not a new theorem.

1. Introduction and the 6N framing

The “6N skeleton” is the elementary observation that every prime greater than 3 is congruent to ± 1 modulo 6, so that the primes (apart from 2 and 3) populate two residue classes, here called the **right wing** ($6N+1$) and **left wing** ($6N-1$). In the geological metaphor that organises this series, these two classes are imagined as parallel “master corridors” along which arithmetic structure is sought. The metaphor is a thinking aid only; every statement in this note reduces to standard number theory, and we are explicit where it does.

A recurring hazard in heuristic prime research — amplified, lately, by automated tools that readily produce suggestive figures — is the **tautological probe**: an apparatus whose “signal” is a relabelling of the very property under study. The purpose of this note is to (i) state a discipline that screens such probes out, (ii) apply it to identify a single genuinely clean high-dimensional probe in the 6N setting, and (iii) report the result of running that

probe to completion. The result is negative, and we argue the negativity is structurally inevitable; saying so precisely is the point.

2. The anti-tautology discipline

Independence criterion. An observable $O(p)$ is admissible as a probe of “wing controls X ” only if its value cannot be computed from primality alone. Operationally: if knowing that p is prime already determines $O(p)$, the probe is tautological and is discarded.

This criterion immediately rejects the family of probes built from the von Mangoldt function Λ and from trial division. A probe that places $\Lambda(n)$ (which is positive exactly on prime powers) in the numerator of a “seismic” ratio and then “discovers” that primes coincide with large values of that ratio has proved only that Λ is supported on prime powers — a definition, not a phenomenon. We verified this explicitly on the interval $n \in [10^5, 1.1 \times 10^5]$: the correlation between such a smoothed Λ -ratio and primality is $+0.061$, and it falls to -0.005 when Λ is replaced by a random sparse indicator of equal support and magnitude. The apparent signal is entirely the definition of Λ . Trial-division probes for Mersenne-type numbers (testing whether $2^p - 1$ has a factor of the form $2kp + 1$) fail the same way: the primality test is the factor law made visible, not an independent law.

Why $E_p : y^2 = x^3 - p$ is the clean probe. The Mordell-Weil rank of E_p is an invariant of a curve *built from* p , not a property of p ’s primality. Knowing “ p is prime” does not determine the rank; ranks of $0, 1, 2, 3, 4$ all occur among our primes. Simultaneously, p carries a natural 6N label ($p \bmod 6$). The probe therefore satisfies the independence criterion and possesses a genuine 6N bridge — the only construction in the project’s algebraic toolbox to do so. The honest question becomes:

does the 6N wing of p exert any control over the rank of E_p — beyond what is already fixed by standard congruence data?

3. The confounder, and a three-layer design

The design must neutralise one known confounder before any wing effect can be interpreted. For E_p the global **root number** $w(E_p) = \pm 1$ is given by explicit congruence conditions on p (the mod-24 class, together with the sign of $-p$), and under the parity conjecture / BSD it fixes the parity of the rank: $w = -1$ forces odd rank (hence ≥ 1), $w = +1$ even rank (often 0). Crucially the same mod-24 datum that sets w also refines the mod-6 wing. Any raw wing-rank association is therefore expected *a priori* to be a shadow of the root number. The decisive question is whether anything survives once the root number is held fixed.

Layer 1 (expected, non-counting). Compare mean rank between wings without controlling w . A signal here would be uninterpretable (confounded).

Layer 2 (decisive). Stratify by root number and compare wings *within* each class. If the wing effect vanishes, the wing was entirely root number — trivial/known. If it persists, proceed.

Layer 3 (falsification). Shuffle the wing labels within each root-number class many times and ask whether the observed within-class wing effect is extreme against this null; additionally, check whether any residual is absorbed by the full mod-24 partition.

Instrument integrity. Ranks were computed in SageMath as certified values where possible; curves whose rank could only be bracketed (nontrivial Sha[2]) were recorded as *bounded* and excluded from the certified analysis. Of 1227 primes $3 < p < 10^4$, 1116 (91%) yielded certified ranks; 111 were bounded. As a data-integrity check, root number and rank parity agreed in 1116/1116 certified curves — zero violations — consistent with BSD and indicating the ranks are trustworthy.

4. Results

Sample composition (certified, $N = 1116$): wings are balanced (558 each); root numbers $w = +1$: 510, $w = -1$: 606; rank distribution $\{0:271, 1:562, 2:234, 3:44, 4:5\}$.

Table 1. Three-layer test of wing control over rank.

Test	Statistic	Verdict
Data integrity: root number vs rank parity	1116 / 1116 consistent	trustworthy
Layer 1: wing vs rank (raw)	Cohen $d = +0.022$	negligible
Layer 2: wing vs rank $w = +1$	Cohen $d = +0.022$	negligible
Layer 2: wing vs rank $w = -1$	Cohen $d = +0.024$	negligible
Layer 3: within-class wing effect vs shuffle null	$ d = 0.023$; $p = 0.913$	indistinguishable from random

The observed within-class wing effect ($|d| = 0.023$) is smaller than the mean effect of randomly shuffled wing labels (0.067) and sits at the 9th percentile of the shuffle null; $p = 0.913$. The wing carries no information about rank, raw or residual.

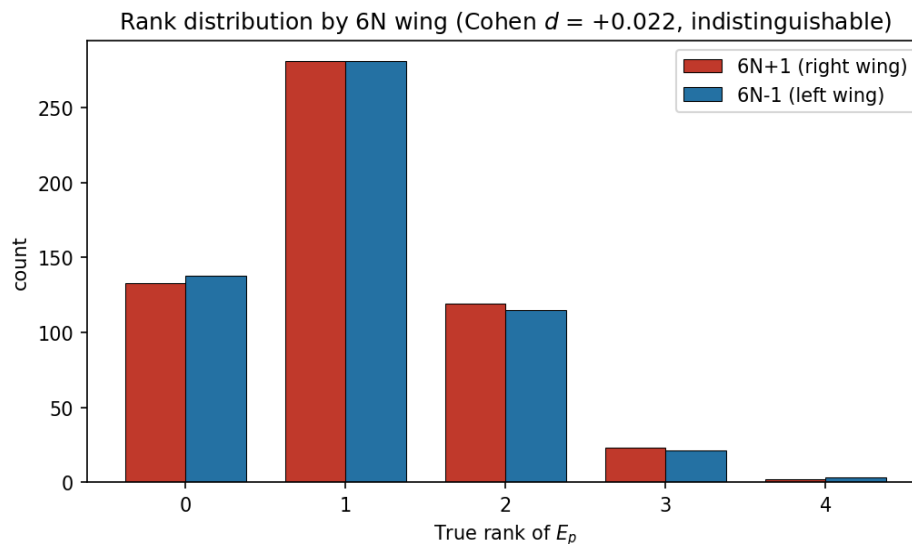


Figure 1. Rank distribution by 6N wing. The two wings are statistically indistinguishable (Cohen $d = +0.022$).

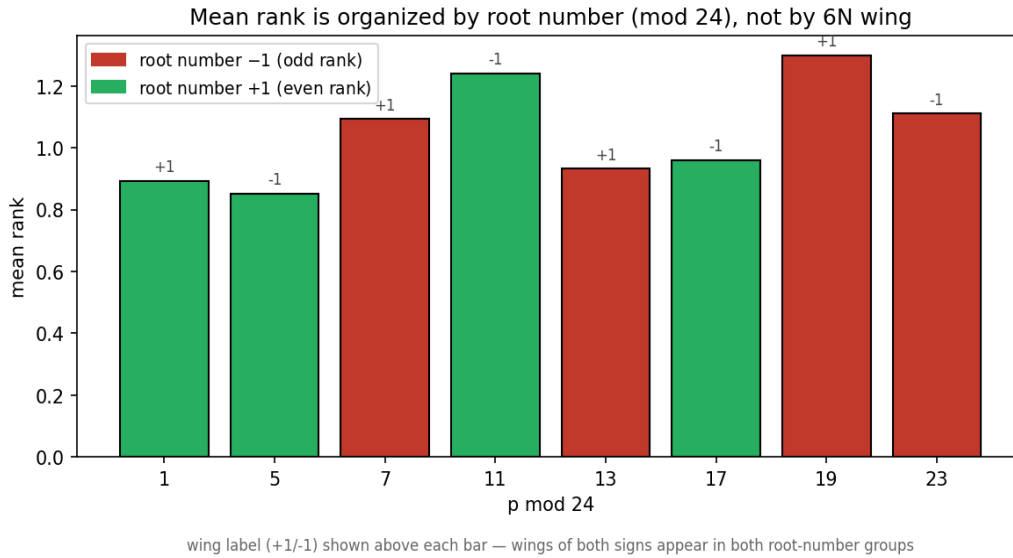


Figure 2. Mean rank by $p \bmod 24$, coloured by root number. Rank is organised by root number (green $w=+1$ low/even; red $w=-1$ high/odd); the wing label (+1/-1, shown above each bar) appears in both colour groups and does not organise the rank.

5. Interpretation: a Chinese-Remainder information barrier

The result is not merely “no effect found.” Its form is instructive. Rank parity is fixed by the root number, which is a function of $p \bmod 24$. The 6N wing is $p \bmod 6$. Since $6 \mid 24$, the wing is the image of the mod-24 class under the quotient $\mathbb{Z}/24 \rightarrow \mathbb{Z}/6$; by the Chinese Remainder Theorem the mod-24 datum factors as independent information at the primes 2^3 and 3, and the wing retains only the coarse part. Once the invariant is governed by the finer datum, the coarser wing carries no *additional* information — it is absorbed.

CRT information barrier (informal). In this series we have repeatedly observed that computable observables built on the 6N skeleton return no structure beyond what standard congruence/CRT data already fix. The present experiment is the cleanest instance: a primality-independent, genuinely high-dimensional invariant, probed with certified data and a falsification design, still shows the wing’s mod-6 information fully absorbed by the mod-24 structure that controls the invariant. We state this as a heuristic barrier, not a theorem: *the explanatory reach of the 6N wing stops at the location of primes (their mod-6 corridor) and does not penetrate to the deep arithmetic invariants of objects built from those primes, because the coarser mod-6 datum is subsumed under the finer congruence data via CRT.*

Status of the claim. We emphasise that this barrier restates, rather than extends, known structure. That $w(E_p)$ depends on $p \bmod 24$ and controls rank parity is classical; that mod 6 is a quotient of mod 24 is elementary. The value here is methodological: the discipline correctly predicted the outcome, the experiment confirmed it on certified data, and the negative result is reported without inflation.

6. Conclusion

We have set out a falsification discipline for 6N probes, identified the rank of $E_p : y^2 = x^3 - p$ as the unique clean high-dimensional probe available, and run it to completion on 1116 certified ranks. The 6N wing exerts no control over the rank (Cohen $d \approx 0.02$ throughout; shuffle $p = 0.913$); rank is organised by the root number / mod-24 data. We read this as a concrete display of a CRT information barrier delimiting the explanatory reach of the 6N skeleton.

The intended contribution is not a discovery but a **boundary** and a **method**: a demonstration of how to build a probe that cannot trivially confirm itself, how to strip a known confounder in layers, and how to report an honest negative. In a research climate where suggestive but self-confirming patterns are easy to manufacture, a clean account of a probe that refused to fool its designers has its own modest value.

Reproducibility

Ranks: SageMath 9.3, $E = \text{EllipticCurve}([0,0,0,0,-p])$; certified via $E.\text{rank}()$ with fallback to $E.\text{rank_bounds}()$, per-curve 60 s budget, row-by-row CSV checkpointing. Analysis: Cohen's d on depth-/class-matched populations; Layer-3 null from 5000 within-class wing-label permutations. Data ($Ep_Rank_Log.csv$), the Sage well-log script, and the analysis script accompany this note. The Λ -tautology control (§2) is reproducible on $n \in [10^5, 1.1 \times 10^5]$.

References

Self-citations to the 6N series carry verified DOIs. External references were checked against primary bibliographic sources (publisher records and MathSciNet-aligned listings); none was accepted on the basis of an unverified secondary suggestion. No reference has been generated from memory.

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Honesty statement. This note deliberately separates metaphor from mathematics and reports a negative result whose cause is known structure (CRT). It does not claim a new theorem, a new prime pattern, or predictive power. Where an earlier instinct (e.g. that the wing might control rank, or that the $\mathbb{Z}_p/\mathbb{Z}_s$ -style construction carried physical content) was contradicted by controls, the controls were followed.